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Applied Statistics

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Abstract

The advent of electronic devices has revolutionized the way we live our lives, with young adults and adolescents at the forefront of this upcoming technological wave. While technology has brought countless benefits to our daily routines, it has also sparked concerns about its potential adverse effects on our health - particularly with regard to sleep habits. This study examines the relationship between screen time and sleep habits, and is designed to provide insights by analyzing data collected from individuals who frequently use electronic devices to determine whether increased screen time is correlated with reduced sleeping hours or not

1 Introduction

The excessive use of electronic devices has become a widespread concern in recent years. The convenience and accessibility of technology have led to increased usage, particularly among young adults and adolescents. However, studies have suggested that this increased screen time may have a negative impact on our health, particularly our sleep. The blue light emitted by electronic devices can disrupt the production of the sleep hormone, melatonin, which can lead to difficulty falling asleep and staying asleep. Additionally, the stimulation of the brain from using electronic devices can make it challenging to unwind and relax, further exacerbating sleep disturbances.

Sleep is essential for many physiological functions, including physical recovery, memory consolidation, and immune system function. The negative effects of sleep deprivation are well-documented, with research indicating that inadequate sleep can lead to a range of adverse health outcomes, including obesity, diabetes, and cardiovascular disease. Given the importance of sleep for our health and well-being, understanding the relationship between screen time and sleep duration is crucial.

This study aims to explore the relationship between screen time and sleep duration, specifically investigating whether increased screen time is associated with decreased sleep duration. By analyzing data collected from individuals who use electronic devices regularly, this study will provide valuable insights into the impact of technology on our sleep patterns. The findings of this study will inform public health policies and guidelines on the use of electronic devices, particularly in terms of recommendations on the amount of screen time people should have to maintain healthy sleep habits. Overall, this study will contribute to our understanding of the complex relationship between technology and our health, highlighting the need for further research in this area.

2 Data Collection

The collection of data was by sending out a Google form to all the students in the Indian Institute of Technology Hyderabad. While designing the form we made sure that the questions are clear and easy to understand, using a mix of question types, including multiple-choice etc. to provide a range of data that can be analyzed. We added demographic questions, such as age to better understand the sample population. The survey consisted of the following questions:

1. What is your age?
2. What device do you spend most of your time on?
3. How do you spend the most time on an electronic device?
4. What is your total screen time per day?
5. Do you constantly check your phone while eating/studying?
6. How many times do you unlock your screen per day?
7. How easy would it be to keep your phone aside for 1 day?
8. Do you have an eye prescription?
9. After how much screen time do you start feeling uncomfortable?
10. How much time do you spend daily exercising/in co-curricular?
11. How much time do you spend daily in academics?
12. How many hours of sleep do you get in a day?
13. How would you rate the quality of your sleep?
14. What time do you usually go to sleep?
15. What time do you usually wake up?
16. How many afternoon naps do you take in a week?
17. Do you think your screen time is affecting your sleep quality?

We used sampling, more specifically volunteer sampling for the collection of data by sending mail to the students. To get the maximum number of responses, we also sent frequent reminders.

The data was collected from nearly 250 people, and it is diverse with data from all years of UG, PG and PhD students. Participants were also asked the number of hours they spend using electronic devices each day and the number of hours they sleep each night. The data collected was securely stored and anonymized to protect participants' privacy.The data collection process was designed to minimize bias and maximize data quality. Overall, the data collection process's aim was to collect reliable and accurate data on screen time and sleep duration to test the hypothesis and determine the relationship between the two.

3 Pre Processing

Preprocessing is a crucial step in any data analysis project, as it helps to ensure that the data is clean, accurate, and in the correct format for analysis.
One of the first things to do is to remove any invalid or incomplete responses. These can include responses that are blank or that contain errors or inconsistencies. By removing these responses, we can ensure that the remaining data is of high quality and can be used for analysis.

Another important step is to check for outliers and remove any extreme values. Outliers can skew the results of the analysis and make it difficult to draw meaningful conclusions. By removing outliers, we can ensure that the analysis is based on reliable data.

Standardizing or normalizing the data can also be helpful, as it allows us to compare different variables on the same scale. This is particularly useful when dealing with variables that have different units or ranges.

If there are missing values in the data, it is important to impute them in a way that avoids bias in the analysis. This can be done using techniques such as mean imputation, median imputation, or regression imputation.

Finally, it is important to remove any unnecessary variables that are not relevant to the analysis. This can help to simplify the analysis and ensure that the results are interpretable. Additionally, checking for multicollinearity and removing any correlated variables can help to avoid issues such as overfitting.

Overall, preprocessing is a critical step in any data analysis project, as it helps to ensure that the results are accurate and reliable. By taking the time to clean and prepare the data, we can ensure that the analysis is based on high-quality data that is appropriate for the research question at hand.

4 Study of Response Variables

Here we'll go into our data and get some inferences from it using graphical representation and also dig into the data and find some statistics .

4.1 Central Tendencies for Screen and Sleep Time

4.1.1 Screen Time

Count	244.000000
Unique	4.00
Top	4 - 7 hours
Frequency	109.00

Table 1: Central Tendencies for Screen Time

The central tendencies are as follows:

1. Count - It refers to the total number of students who have responded
2. Unique - It refers to the number of unique values in the column. This is often used to determine the cardinality of categorical data.
3. Top - It refers to the most frequent value in the column.
4. Frequency - It refers to the number of occurrences of the top value

4.1.2 Sleep Time

Count	244.000000
Mean	6.856557
Std	1.428659
Min	0.000000
25%	6.000000
50%	7.000000
75%	8.000000
Max	10.000000

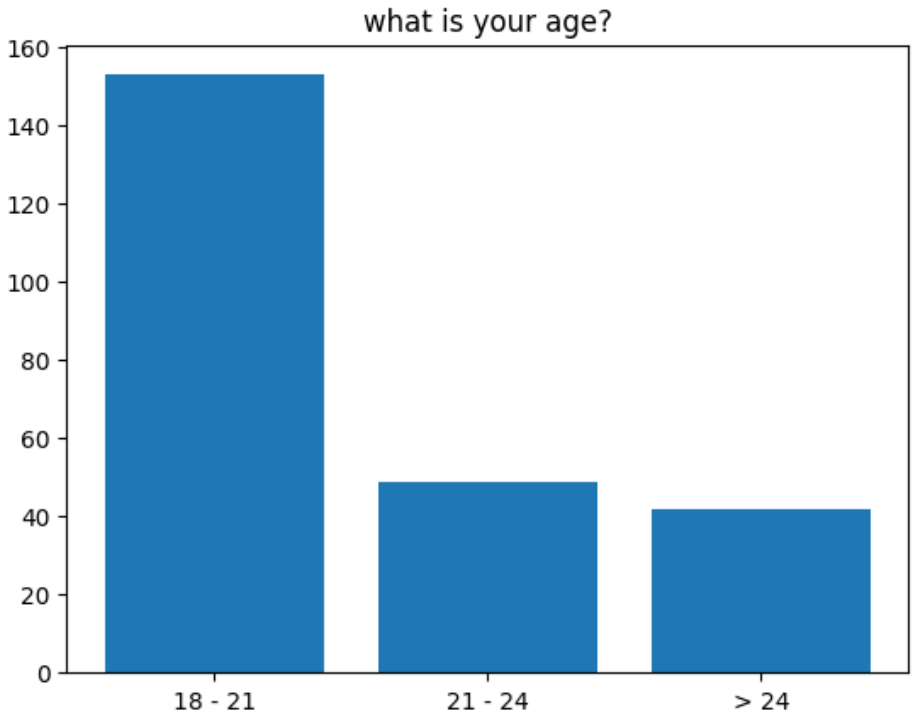
Table 2: Central Tendencies for Sleep Time

The central tendencies are as follows:

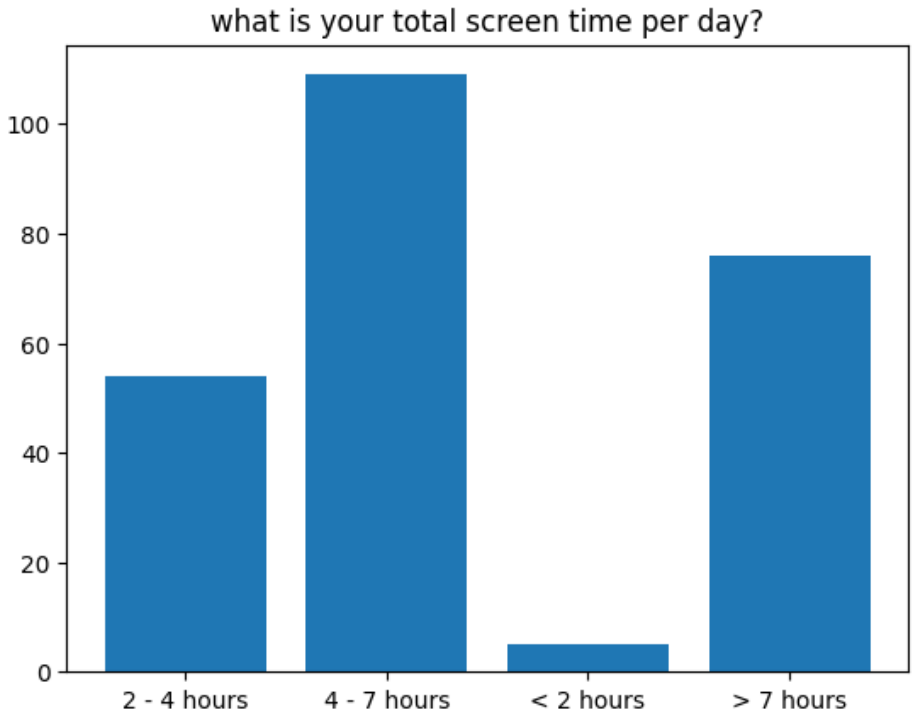
1. Count - It refers to the total number of students who have responded
2. Mean - It refers to the mean duration of sleep hours
3. Std - It refers to the standard deviation of the sleep hours surveyed, i.e., the spread of the values
4. Min - It refers to the minimum number of hours someone has responded with
5. 25% - It refers to the value at the 25th percentile of the column, which separates the bottom 25% of the values from the top 75% of the values
6. 50% - It refers to the value at the 50th percentile, also known as the median, which separates the bottom 50% of the values from the top 50% of the values.
7. 75% - It refers to the value at the 50th percentile, also known as the median, which separates the bottom 75% of the values from the top 25% of the values.
8. Max - It refers to the maximum number of hours in the response

4.2 Data Visualization

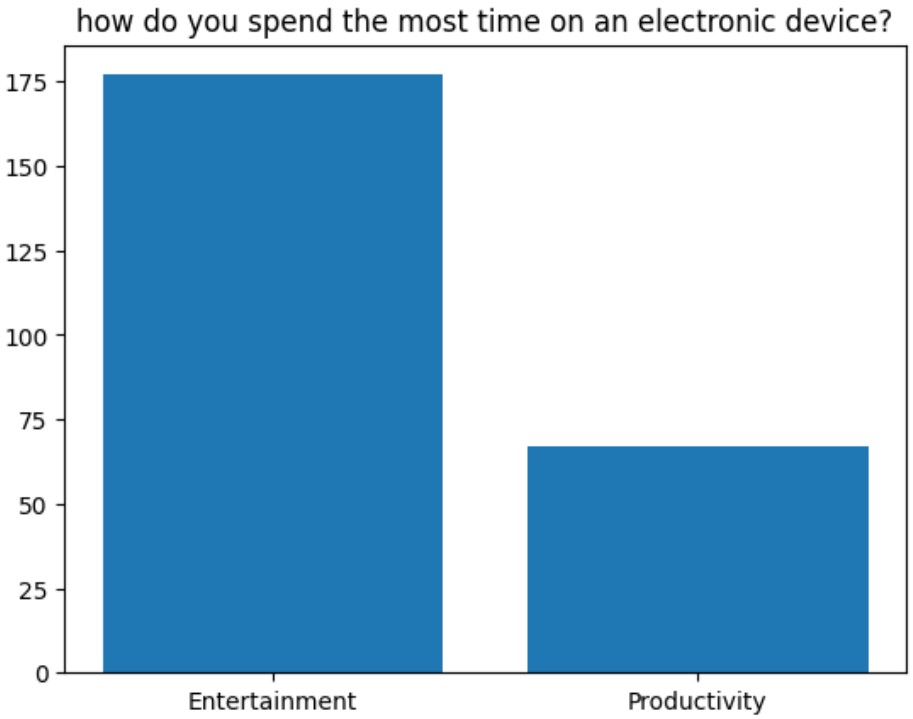
We visualize the data we have obtained from the data collections. Firstly we see the age range of the obtained data



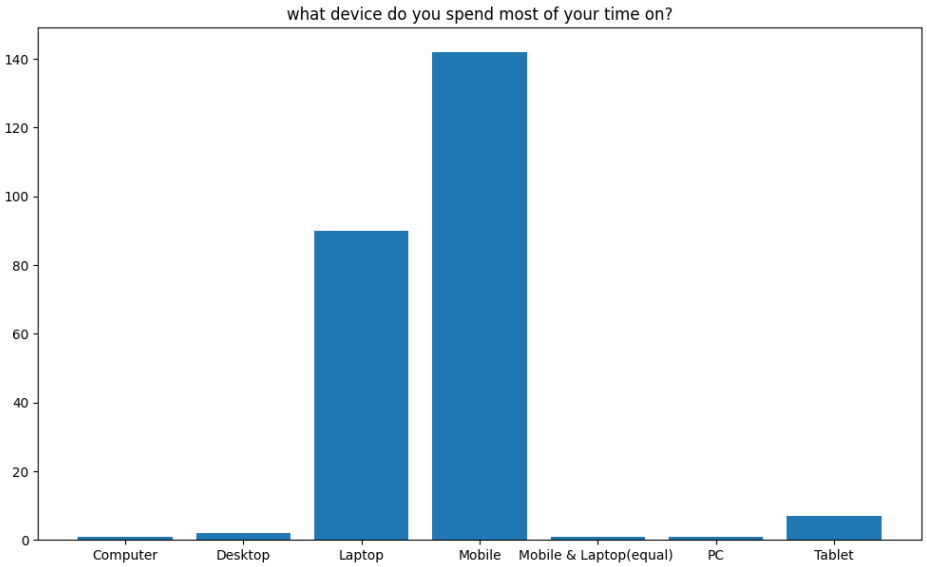
Most of our data come from the age range of 18-21.We got a total of 153 people in that range. Next we look into the screen time of population.



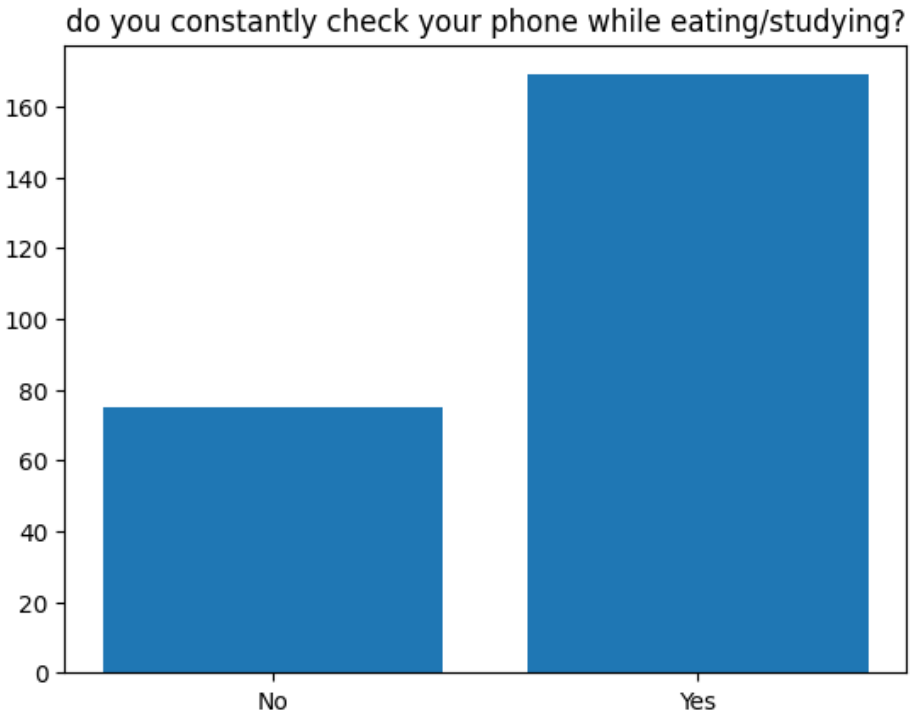
We see that most of the people have around 4-7 hours of screen time and the second highest being more that 7 hours. Let's now look into where the population spend time most of their screen time.



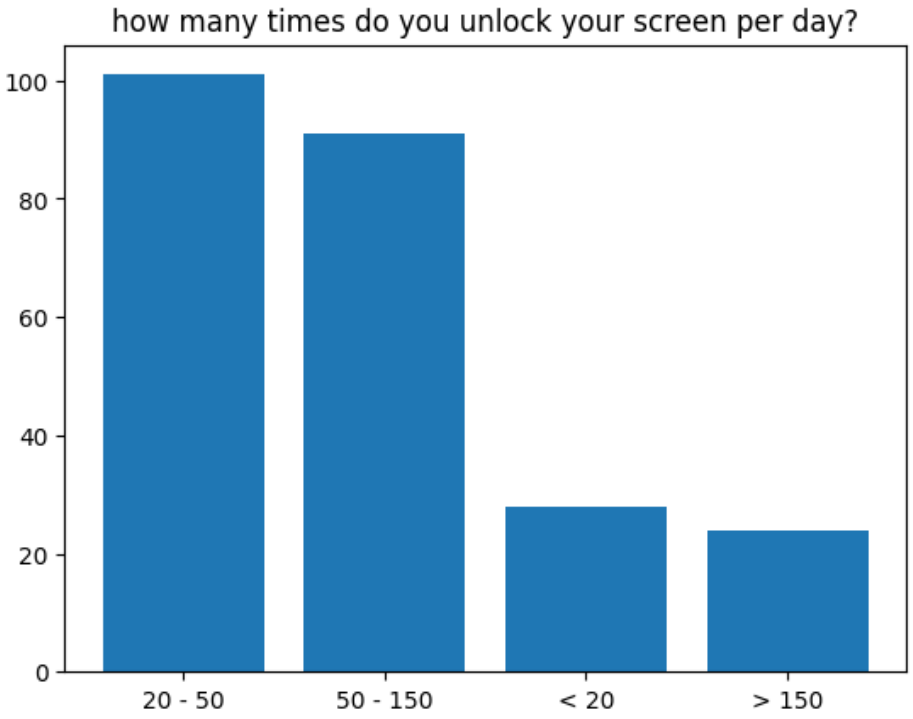
Here, the majority is entertainment. Which device do they use the most?



We see the mobile is used the most.

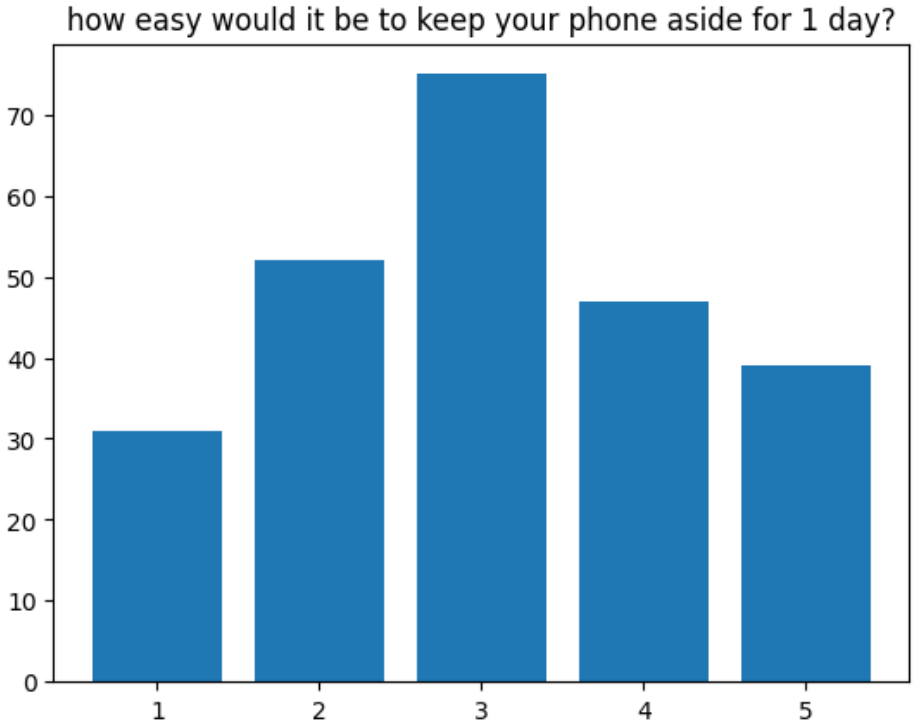


We observe that 69.1% people check their phone while eating/studying.

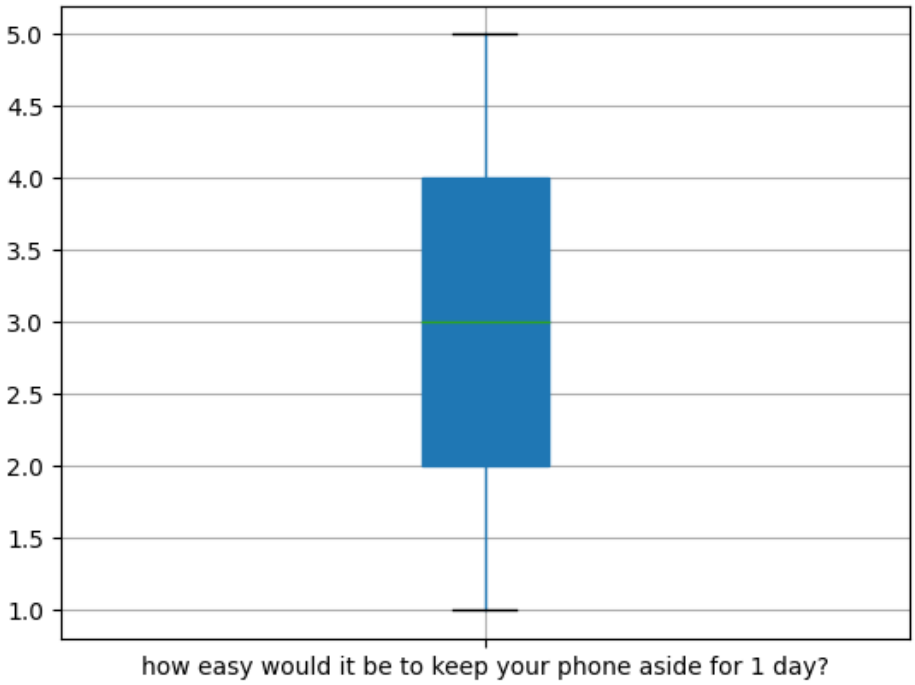


We observe that the relative percentages of responses in the category follow 11.7%, 42%, 36.4% and 9.8% in increasing order of number of unlocks. We see that the highest percentage of students unlock their phone 20-50 times a day.

The question "How many of you would be able to keep your phones aside for a day?" is asking about the willingness or ability of people to disconnect from their mobile devices for a day.



To understand the results of this question better, we can use a box plot.



A box plot is a visual representation of the distribution of a dataset that displays the median, quartiles, and outliers. In this case, we can use a box plot to show the distribution of responses to the question about keeping phones aside for a day.

To create the box plot, we would first need to collect the data on people's responses, either through a survey or other means. Once we have the data, we have used a plotting library in Python: Matplotlib to create the box plot. It displays the following elements:

1. The median of the data, which represents the value that separates the top 50% from the bottom 50% of the responses.
2. The interquartile range (IQR), which is the range between the first and third quartiles of the data. The first quartile (Q1) represents the value that separates the bottom 25% of the responses from the top 75%, and the third quartile (Q3) represents the value that separates the top 25% from the bottom 75% of the responses.
3. The whiskers, which represent the range of the data within 1.5 times the IQR from the first and third quartiles. Any values outside this range are considered outliers and are displayed as individual points.
4. The box, which represents the range of the data between the first and third quartiles.
5. Any outliers, which are displayed as individual points outside the whiskers.

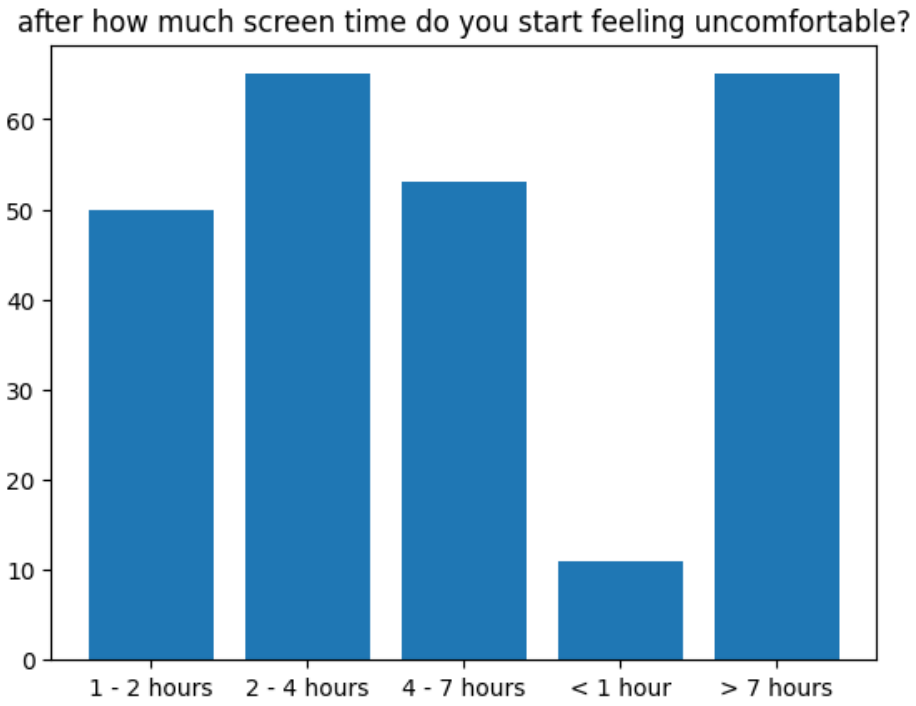
Interpreting the box plot can give us insights into the distribution of responses to the question about keeping phones aside for a day. Here,

Median	3
Q1	2
Q3	4
IQR	2

Table 3: Box Plot Inferences

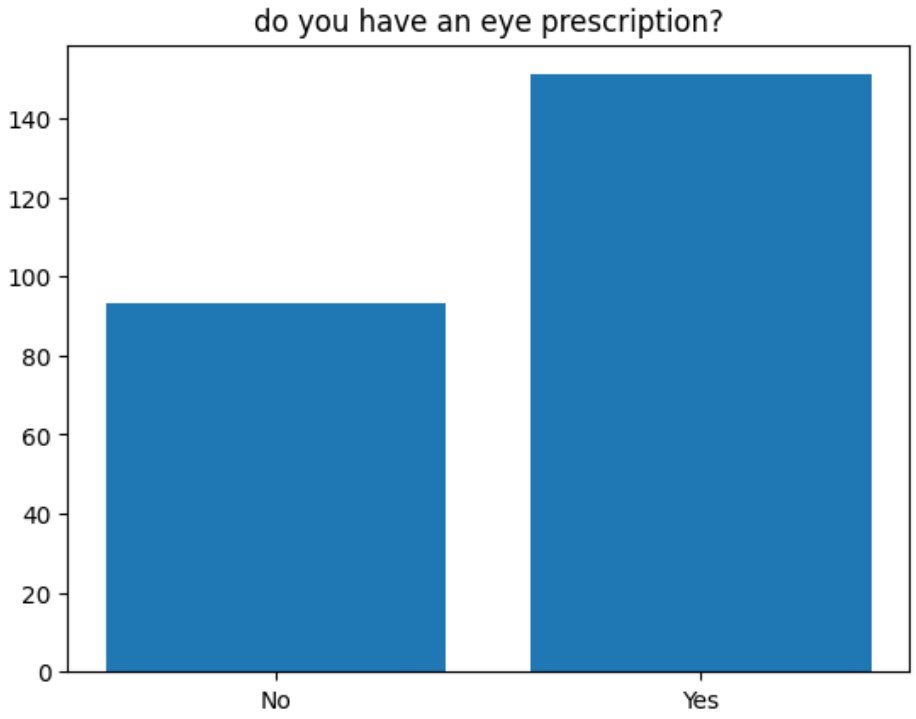
Overall, a box plot can be a useful tool for visualizing the distribution of responses to a question like this, and can help provide insights into the attitudes and behaviors of people with respect to their mobile devices.

Now let's look into how it screen time affects the eyes.

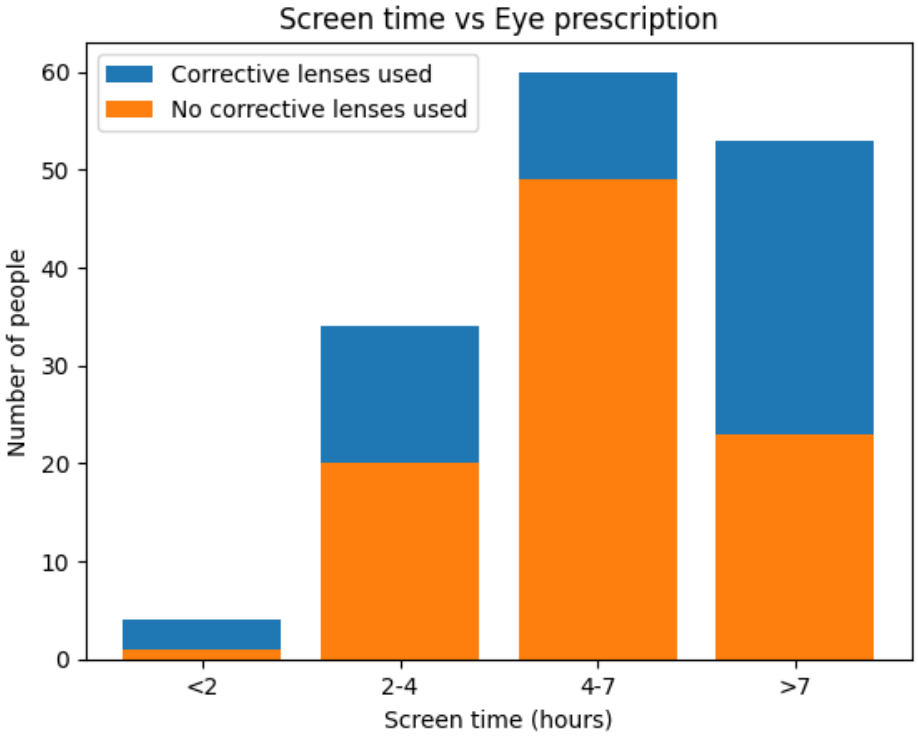


We observe that most people feel uncomfortable after 2-4 hours of screen time, followed closely by >7 hours of screen time.

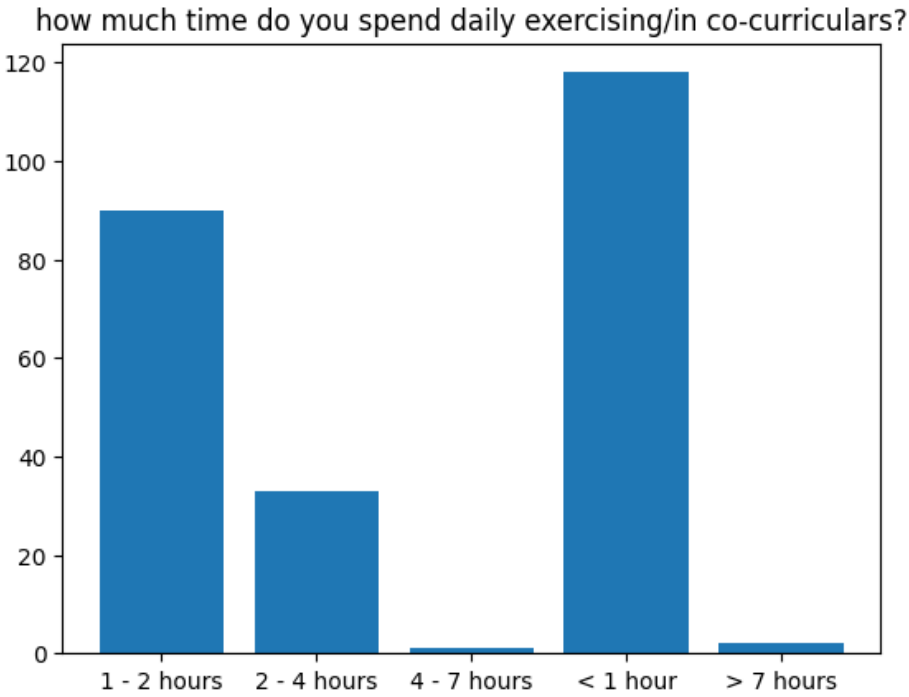
To understand the correlation between eye damage and screen time, we could look at the prevalence of eye prescriptions in individuals who spend a lot of time on screens versus those who do not.



We also see the visualization of the number of people who use corrective lenses in comparison to their screen time.

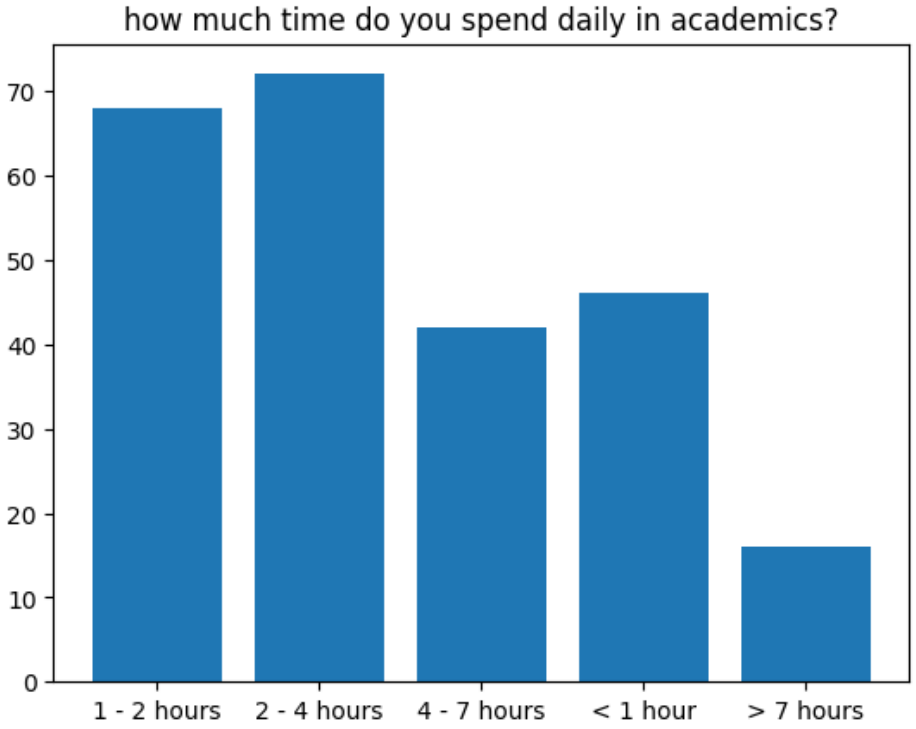


Daily exercise and co-curricular activities are important for a number of reasons, including physical health, mental health, and social development.



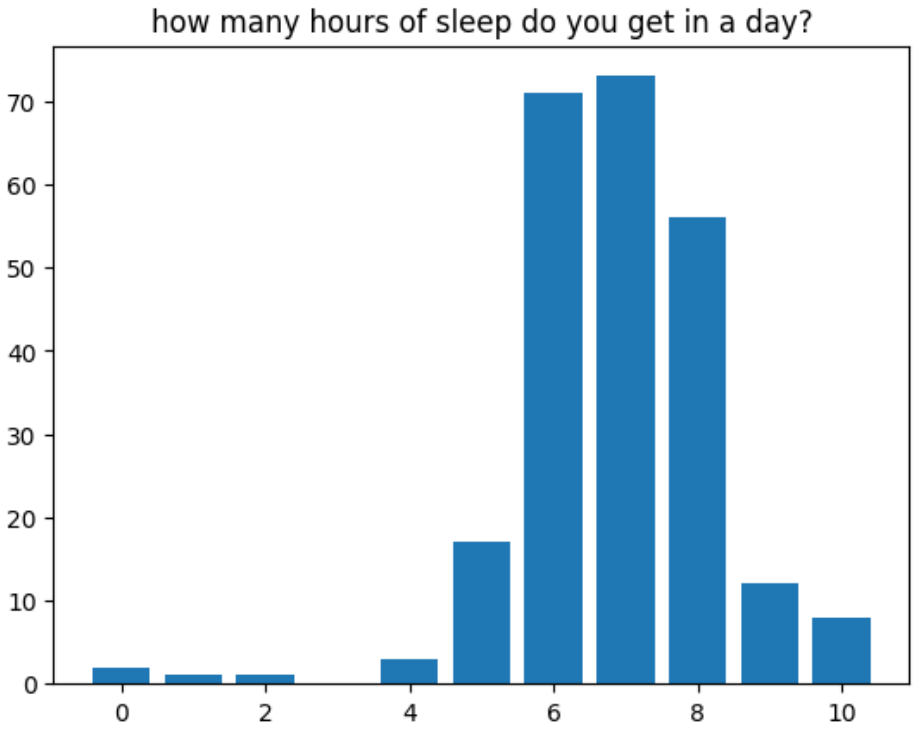
We observe the demographic and see that 50% of them spend less that 1 hour for the same, with 36.4% spending 1-2 hours on daily excercise/Co-Curriculars.

The question "how much time do you spend daily on academics" is a valid question in a survey because it provides valuable information about the level of academic engagement and workload of the respondents. This information can be important for understanding factors that may impact academic performance, such as time management, study habits, and the availability of resources.



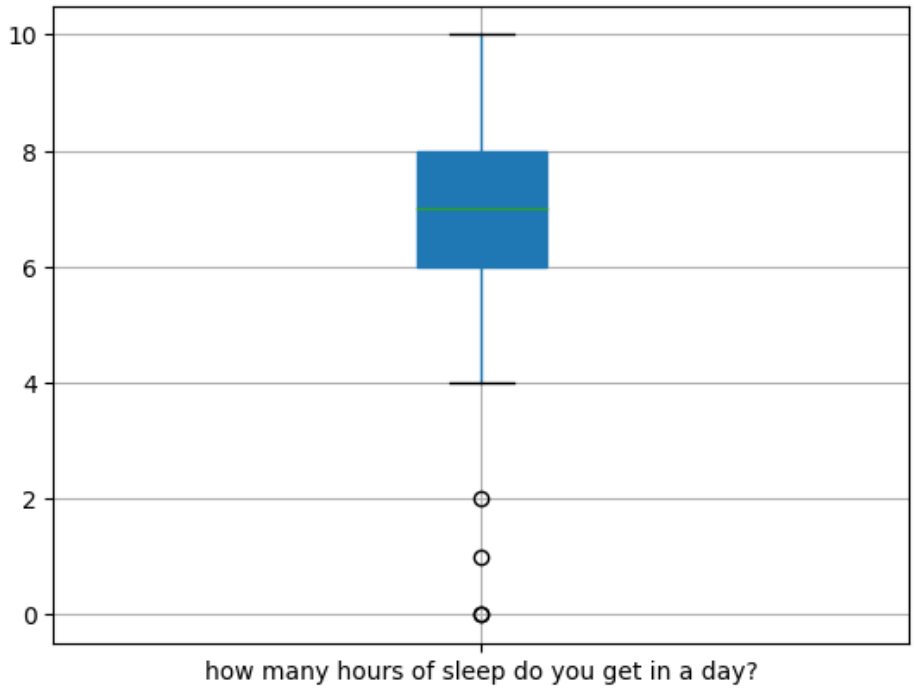
We observe that 2.88% of the students spend between 1 and 2 studying per day and 28% of the students spend between 2 and 4 hours studying per day.

Data on sleeping times and habits is important as it can provide insight into the sleep patterns of a population, which can be useful in understanding factors that may impact sleep, such as lifestyle, work schedules, and health conditions. Let's now see the data visualization on sleeping times and habits.



We gauge the average sleep time of the entire demographic and observe that 29.8% responded with 7 hours of sleep, 29.1% with 6 hours, and 22.6% responded with 7 hours of sleep. We compare this with a 2019 study conducted by the US-based firm Fitbit across 18 countries that declared the average sleep time of Indian citizens as 7 hours and 1 minute.

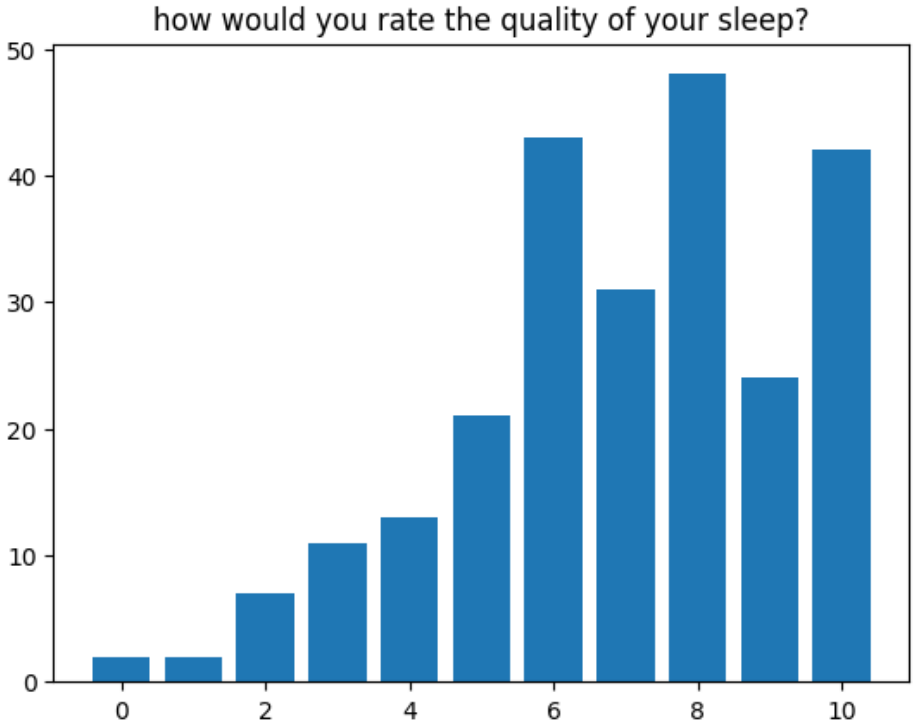
The box plot for the same results in



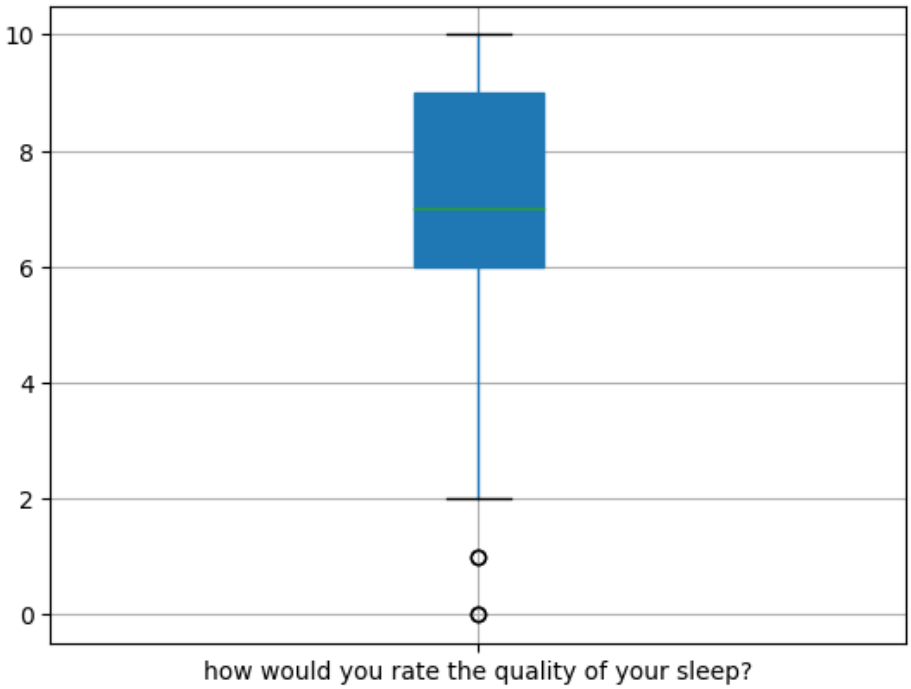
Median	7
Q1	6
Q3	8
IQR	2

Table 4: Box Plot Inferences

Quality of sleep refers to how restful, restorative, and refreshing an individual's sleep is perceived to be. Poor quality sleep can result in a range of negative consequences, including fatigue, irritability, and impaired cognitive function. In addition, chronic poor quality sleep has been linked to a variety of health problems, including obesity, diabetes, cardiovascular disease, and depression.



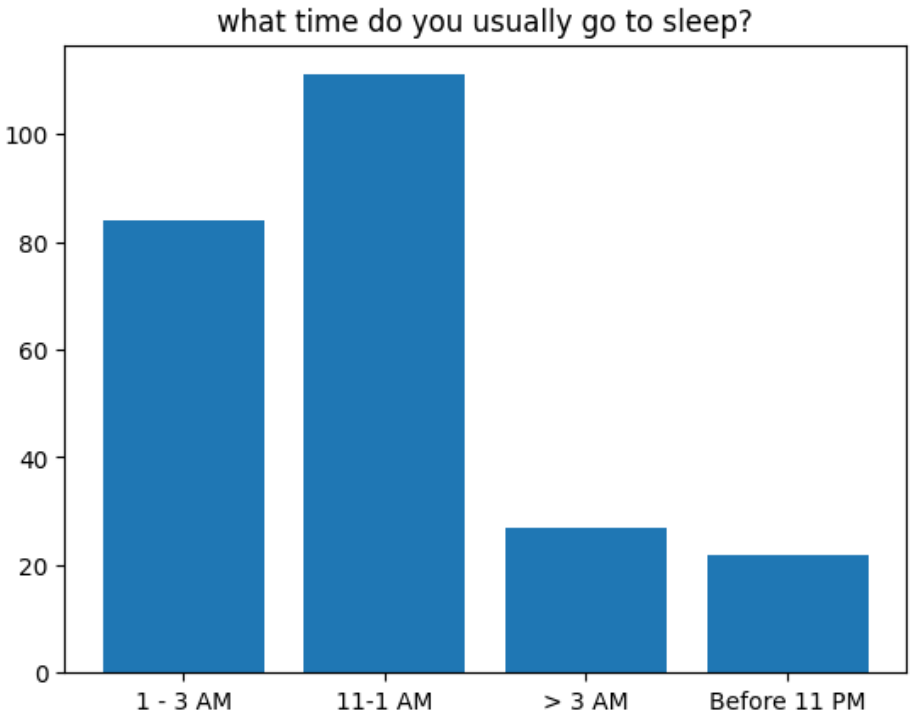
We observe that 20.4% rated their quality of sleep as an 8/10. The Box plot can be seen.



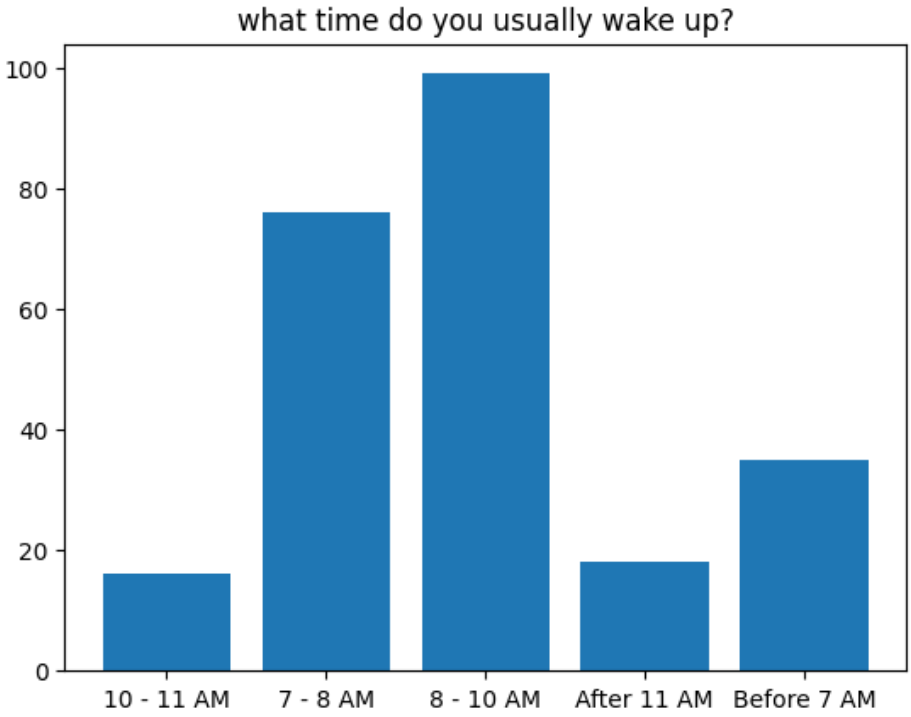
Median	7
Q1	6
Q3	9
IQR	3

Table 5: Box Plot Inferences

"What time do you usually go to/wake up from sleep?" is a valid question for a survey focused on sleep habits and patterns. The response to this question can provide valuable information on the average bedtime of the respondents, which can be important for understanding their sleep duration and quality. Additionally, this question can help identify any patterns or trends in bedtime preferences.



We observe that around 44.7% of the demographic go to sleep between 11 PM and 1 AM, closely followed by 34.8% going to sleep between 1 AM and 3 AM.

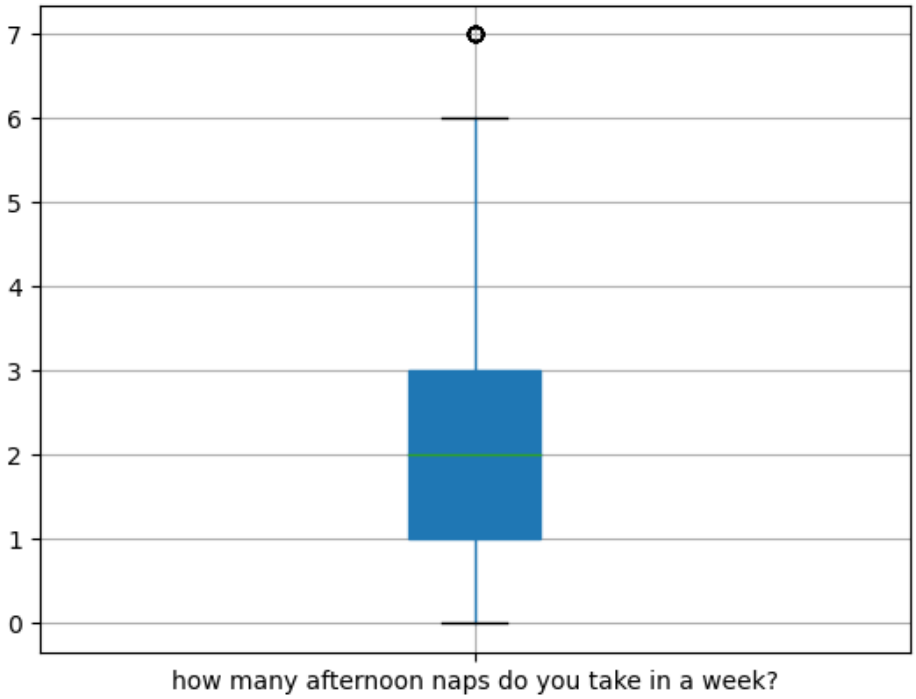


We observe that around 40.6% of the demographic wake up between 8 AM and 10 AM, followed by 30.5% waking up between 7 AM and 8 AM.

Afternoon naps, also known as "power naps," have been shown to have a range of benefits, including improved cognitive function and memory consolidation, reduced stress, and increased alertness. Further, they also contribute to the study of sleep habits among the demographic.



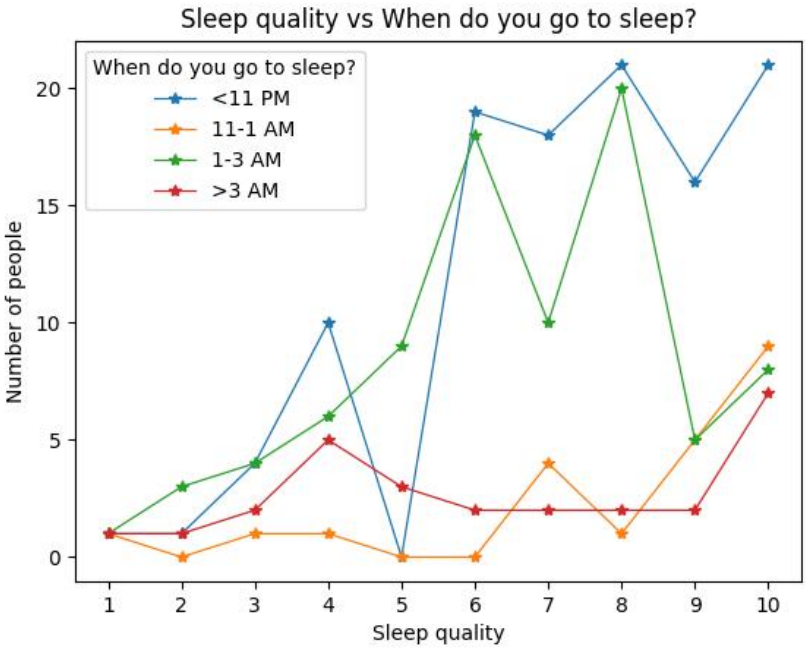
We observe a mean value of 2.217213, with a standard deviation of 1.968358. Considering the Box plot,



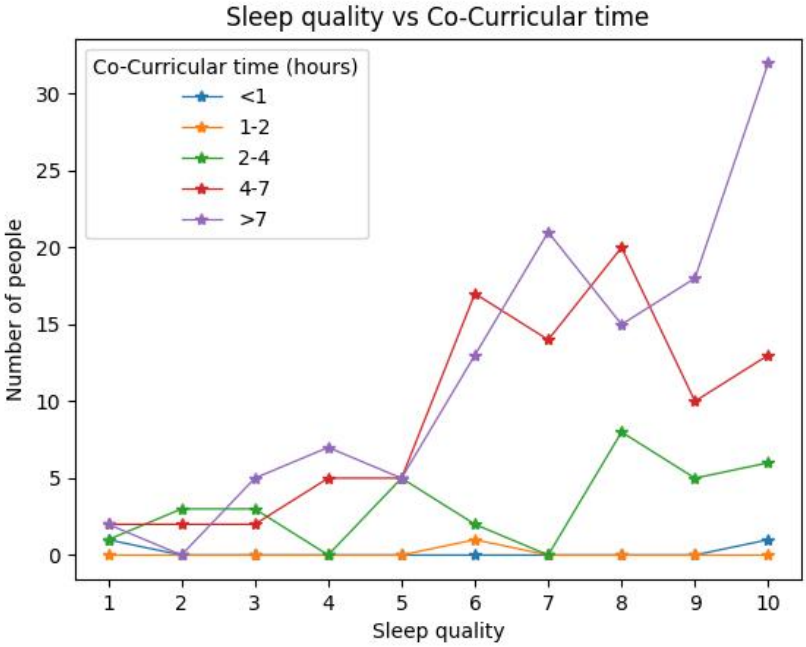
Median	2
Q1	1
Q3	3
IQR	2

Table 6: Box Plot Inferences

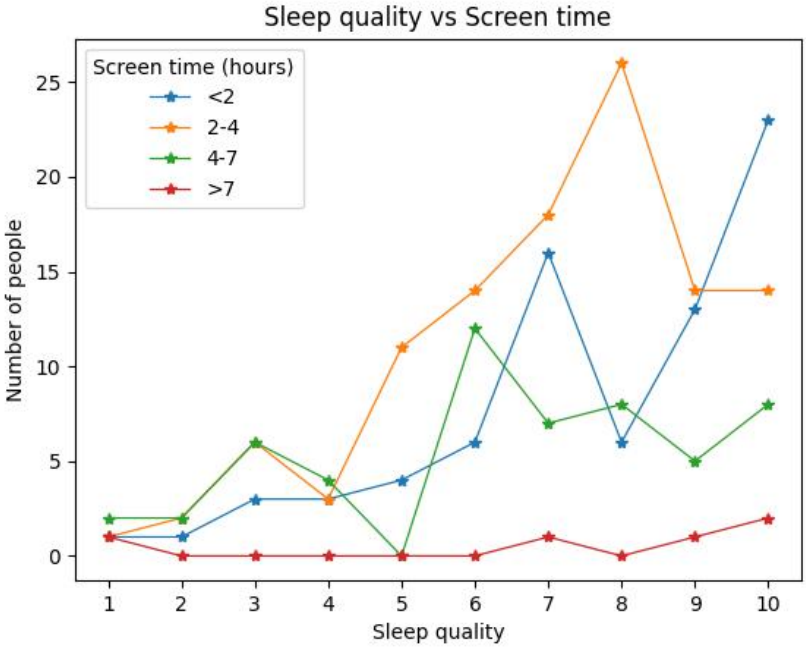
More graphs,



Here each coloured lines represent the time at which they sleep at for example red line indicates the people who sleep after 3 am and green line indicates those who sleep between 1 am to 3 am.We can see that we have high number people who sleep early have better sleep quality.

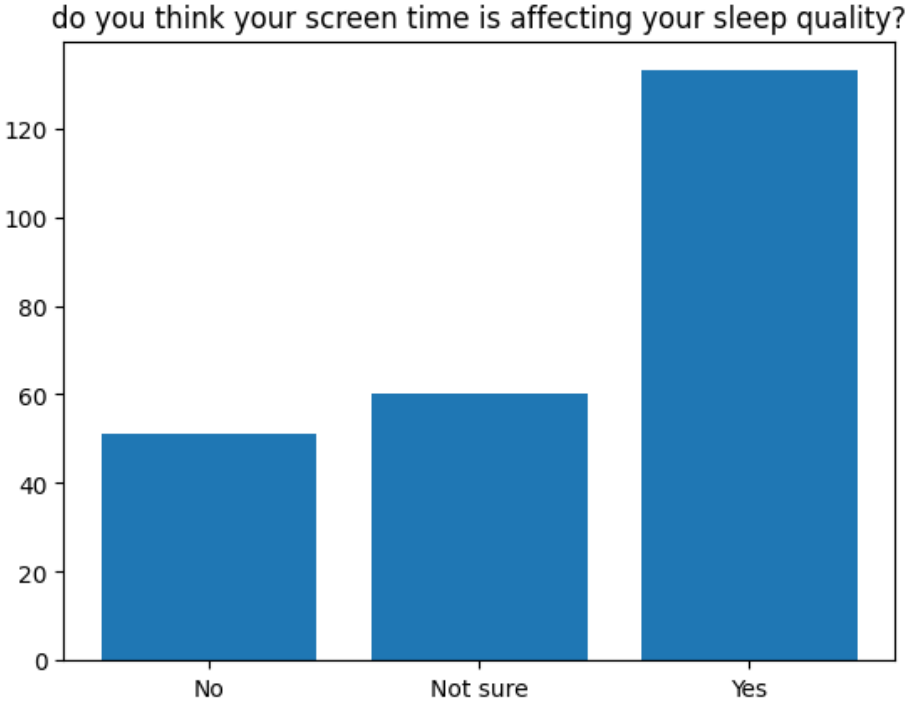


Here coloured lines represent the time spent on Co-Curricular activities. We can see that there are more number of people who spend time on Co-Curricular activities have a very good sleep quality.



Here the coloured lines represent the screen time.We can see that we have high number of people who have less screen time have better sleep quality.

Finally, we even asked our participants about their opinion on 'does screen time affect sleep' and this is what we got.



We see that most of our participants think that screen time affects their sleep.

5 Hypothesis Testing

Hypothesis testing is a statistical inference on a dataset through which we can infer our hypothesis from the data given in the data. Hypothesis testing usually starts with identifying a hypothesis, specifying what our Null Hypothesis(H_0) and Alternate Hypothesis(H_a), deciding the test hypothesis and check for either of p-values is statistically significant or not(p-value approach) or whether the Null hypothesis falls under rejection region or not(Rejection Region approach)

Set of hypothesis we have tested -

1. **Hypothesis 1:** People who have an eye prescription have an average higher screen time(7 hours).
2. **Hypothesis 2:** People who have more screen time spend have an average time of 2 hours of exercises/co curricular activities.
3. **Hypothesis 3:** People with more screen time experience poorer sleep quality.
4. **Hypothesis 4:** People with more screen time tend to sleep later.

We now look into each Hypothesis in greater detail.

5.1 Hypothesis 1

We have a sample of people who have answered 'Yes' for having an eye prescription from the population data collected. We now see the trend in screen time.

5.1.1 Hypothesis Formulation

- Null Hypothesis (H_0) - Mean of the above sample is less to 7.
- Alternate Hypothesis (H_a) - Mean of the above sample is greater than or equal to 7.

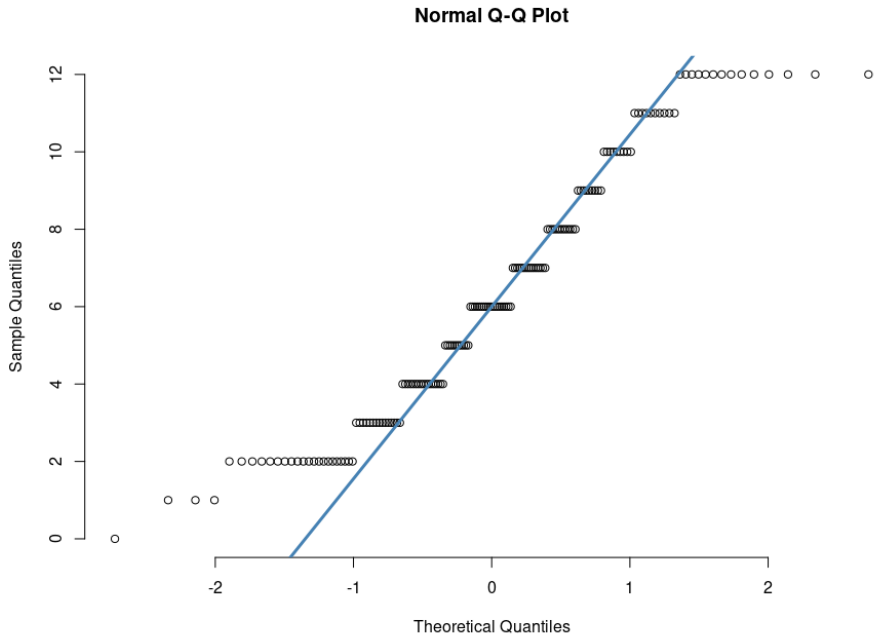
5.1.2 Central Tendencies of the sample

For the above sample,

- Mean($\bar{x}$) = 6.26,
- Standard Deviation(σ_x) = 3.30,

5.1.3 Selecting Test Statistic

On checking whether the data follows normal distribution, we plot the data in a **QQ Plot** and visualise whether it follows normal distribution or not



We see that the plot is not following a straight line although we can see that there is a step wise decrease along the diagonal. It is because the data points we have are discrete and not continuous, it seems as if the data is not following normal. Data follows normal because data is discrete and is diagonally decreasing in step wise intervals.

From here, since the population variance is a unknown quantity and we only have a sampled estimated, choosing **right tailed t test** as test statistic is ideal. On performing right tailed t test to the hypothesis we get the following:

```
        One Sample t-test

data:  dfHyp1
t = -2.7866, df = 155, p-value = 0.997
alternative hypothesis: true mean is greater than 7
95 percent confidence interval:
 5.825065      Inf
sample estimates:
mean of x
 6.262821
```

5.1.4 Inferences

From the above Hypothesis testing, **p-value** is 0.997 which is $\geq \alpha$ (=0.05). Hence, we fail to reject the Null Hypothesis, that is, we lack evidence to reject that average screen time is less than 7 hours for those who have an eye prescription. From the confidence intervals, we can infer that average screen time for people with eye prescription is above 5.8 hours with 95%

confidence. Hence, from the hypothesis testing, we can conclude that people who have an eye prescription are more probable to have higher screen times.

5.2 Hypothesis 2

We have a sample of people who have answered more than 4 hours of their spending in screen time from the population data collected. We now see the trend in hours spent in exercise/co curricular activities

5.2.1 Hypothesis Formulation

- Null Hypothesis (H_0) - Mean of the above sample is less to 2 hours.
- Alternate Hypothesis (H_a) - Mean of the above sample is greater than or equal to 2 hours.

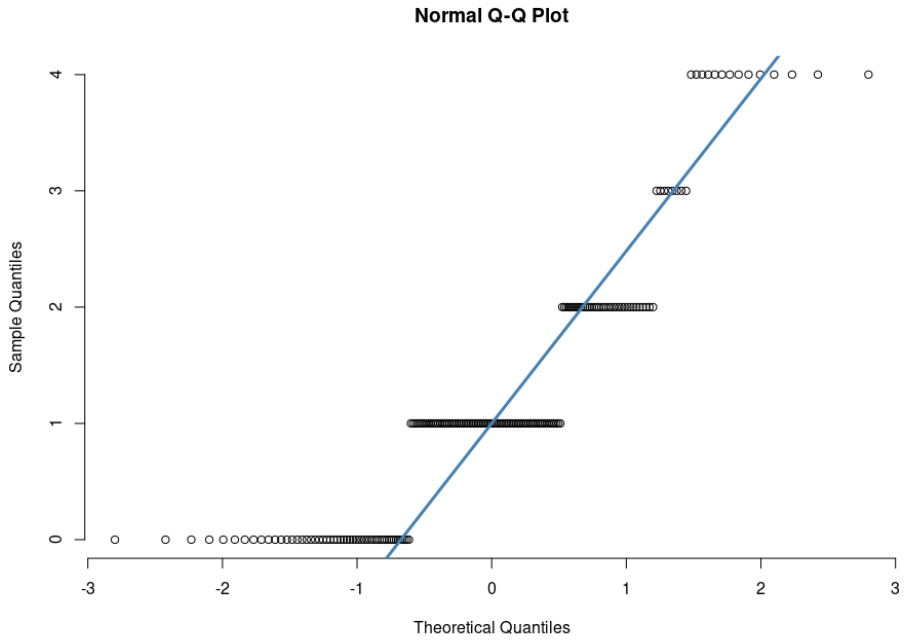
5.2.2 Central Tendencies of the sample

For the above sample,

- Mean($\bar{x}$) = 1.215,
- Standard Deviation(σ_x) = 1.109,

5.2.3 Selecting Test Statistic

On checking whether the data follows normal distribution, we plot the data in a **QQ Plot** and visualise whether it follows normal distribution or not



We see that the plot is not following a straight line although we can see that there is a step wise decrease along the diagonal. It is because the data points we have are discrete and not continuous, it seems as if the data is not following normal. Data follows normal because data is discrete and is diagonally decreasing in step wise intervals.

From here, since the population variance is a unknown quantity and we only have a sampled estimated, choosing **right tailed t test** as test statistic is ideal. On performing right tailed t test to the hypothesis we get the following:

```
        One Sample t-test

data:  df2
t = -9.8723, df = 194, p-value = 1
alternative hypothesis: true mean is greater than 2
95 percent confidence interval:
 1.08403     Inf
sample estimates:
mean of x
 1.215385
```

5.2.4 Inferences

From the above Hypothesis testing, **p-value** is 1 which is $\geq \alpha$ (=0.05). Hence, we fail to reject the Null Hypothesis, that is , we lack evidence to reject that average hours spent in exercise / co curricular activities less than 2. From the confidence intervals, we can infer that average screen time for people with eye prescription is above 1.08 hours with 95% confidence. Hence, from the Hypothesis testing, we can conclude that people having over 4 hours of screen time are more probable to spend less than 2 hours in exercises / co curricular activities.

5.3 Hypothesis 3

We use the **Spearman rank correlation** to check the dependency between the screen time and sleep quality variables.

5.3.1 Hypothesis Formulation - Spearman correlation

To perform this, we being by assigning ranks to each of the observational values of screen time variable and sleep quality variable, 1 depicting highest values, 2 for second highest and so on. We then take the differences in the rank and observe the correlation in terms of squares of differences and Spearman rank coefficient. The range of the Spearman's rank correlation coefficient is $[-1, 1]$ Spearman's rank correlation coefficient, ρ can be calculated as follows -

$$\rho = 1 - \frac{6 \times \Sigma d_i^2}{n(n^2 - 1)} \tag{1}$$

Where,

- ρ is the Spearman's rank Correlation coefficient.
- d_i^2 is the squares of differences in the rank.
- n is the number of observations.

Hypothesis can be written mathematically as:

- Null Hypothesis (H_0) - "Screen time and sleep quality are independent", $\rho = 0$
- Alternate Hypothesis (H_a) - "Screen time and sleep quality are correlated", $\rho \neq 0$

5.3.2 Central Tendencies of the rank difference

For the above sample,

- Correlation coefficient (ρ) = 0.0119,
- Mean difference between screen time and sleep quality($\bar{d_i}$) = 0.666,
- Standard deviation of differences between screen time and sleep quality (σ_{d_i}) = 0.889

5.3.3 Selecting Test Statistic

Here, we have computed ρ and we take the following test statistic S - which is the sum of squared differences of each observation.

```
        Spearman's rank correlation rho

data:  st2 and sq
S = 1278237, p-value = 0.8673
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
0.01195155
```

5.3.4 Inferences

From the above Hypothesis testing, **p-value** is 0.8673 which is $\geq \alpha$ (=0.05). Hence, we fail to reject the Null Hypothesis, that is, there is **insufficient evidence** to conclude that there is a significant relationship between screen time and sleep quality because the correlation coefficient is close to zero.

5.4 Hypothesis 4

We use the **Spearman rank correlation** to check the dependency between the screen time and bed-time variables.

5.4.1 Hypothesis Formulation - Spearman correlation

To perform this, we being by assigning ranks to each of the observational values of screen time variable and bed-time variable, 1 depicting highest values, 2 for second highest and so on. We then take the differences in the rank and observe the correlation in terms of squares of differences and Spearman rank coefficient. The range of the Spearman's rank correlation coefficient is $[-1, 1]$ Spearman's rank correlation coefficient, ρ can be calculated as follows -

$$\rho = 1 - \frac{6 \times \Sigma d_i^2}{n(n^2 - 1)} \tag{2}$$

Where,

- ρ is the Spearman's rank Correlation coefficient.
- d_i^2 is the squares of differences in the rank.
- n is the number of observations.

Hypothesis can be written mathematically as:

- Null Hypothesis (H_0) - "Screen time and bed-time are independent", $\rho = 0$
- Alternate Hypothesis (H_a) - "Screen time and bed-time are correlated", $\rho \neq 0$

5.4.2 Central Tendencies of the rank difference

For the above sample,

- Correlation coefficient (ρ) = 0.1351,
- Mean difference between screen time and bed-time($\bar{d}_i$) = 0.539,
- Standard deviation of differences between screen time and bed-time (σ_{d_i}) = 1.032

5.4.3 Selecting Test Statistic

Here, we have computed ρ and we take the following test statistic S - which is the sum of squared differences of each observation.

```
        Spearman's rank correlation rho

data:  st and hs
S = 2622175, p-value = 0.02845
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.1351278
```

5.4.4 Inferences

From the above Hypothesis testing, **p-value** is 0.028 which is $\leq \alpha$ (=0.05). Hence, we reject the Null Hypothesis, that is, there is **sufficient evidence** to conclude that there is a significant relationship between screen time and bed-time because the correlation coefficient is significantly different from zero. From value of ρ, we can see that there is a positive correlation for screen time and bed-time. From this, we can statistically infer that people who spend more time in front of a screen are likely to have a delayed bed-time.

6 Contributions

- Sai Prateek Katkam (ES20BTECH11024) - Code and Data
- Vibhavasu (EP20BTECH11015)- Code and Data
- Gowri Govindaraj (EP20BTECH11007)- Report
- A Anish (ES20BTECH11001) - Report
- V Samyuktha (MS20BTECH11024)- Hypothesis Testing
- B Vishal Datta (ES20BTECH11009)- Hypothesis Testing, Report
- B Varsha Hari Priya (EP19BTECH11024)- Presentation
- Rajkumar (ES20BTECH11032)- Presentation
- Yash Khedkar (ES20BTECH11033)- Presentation